



北京大学

本科生毕业论文

题目: **Householder 变换矩阵稳定性
定性的 Lean 形式化证明**

**Formalization of the Stability
Proof for the Householder
Transformation Matrix in Lean**

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摘要

QR 分解是数值线性代数中的基础问题，在科学计算与工程应用中具有广泛用途。Householder 变换作为求解 QR 分解的重要方法，其数值稳定性直接影响算法的可靠性。随着形式化验证技术的发展，利用定理证明系统对数值算法进行严格验证，已成为提高计算结果可信性的重要研究方向。本文以 Householder 变换的稳定性分析为对象，基于 Lean 定理证明系统，对相关数学结论进行了形式化证明。论文首先对浮点误差模型以及 Householder 变换中的关键计算过程进行了形式化描述，在此基础上完成了稳定性定理的证明，并给出了误差界的具体常数。研究表明，在合理的机器精度条件下，计算误差能够随向量维度保持线性增长，从而保证算法具有良好的数值稳定性。本文工作展示了 Lean 在数值线性代数算法验证中的应用能力，为后续数值算法的可信形式化研究提供了参考。

关键词：数学形式化，Lean

ABSTRACT

QR decomposition is a fundamental problem in numerical linear algebra with wide applications in scientific computing and engineering. As a key method for computing QR decompositions, Householder transformations play an important role in ensuring numerical stability. With the development of formal verification, Lean has become an effective tool for rigorously analyzing and certifying numerical algorithms. This paper presents the formalization of the stability of Householder transformations using the Lean prover. This project formalized the floatpoint error framework and the essential computational steps involved in Householder transformations, and established a formal proof within Lean. Based on this formalization, explicit error bounds are derived, showing that under reasonable machine precision assumptions, the computational error grows linearly with the vector length. This result confirms the numerical stability of the algorithm and demonstrates the effectiveness of Lean in the formal verification of numerical linear algebra proofs, providing a foundation for further work on trustworthy and machine-checked numerical analysis.

KEYWORDS: Formalization, Lean